

Q1:

1) Show that $m(a+bX) = a + b \times m(X)$

$$m(X) = \frac{1}{N} \sum_{i=1}^N x_i, \text{ let } y = a + bX$$

$$m(y) = \frac{1}{N} \sum_{i=1}^N y_i = \frac{1}{N} \sum_{i=1}^N (a + bx_i)$$

$$\frac{1}{N} \sum_{i=1}^N (a + bx_i) = \frac{1}{N} \left[\sum_{i=1}^N a + \sum_{i=1}^N bx_i \right] = \frac{1}{N} \left[N \times a + b \times \sum_{i=1}^N x_i \right]$$

$$\frac{1}{N} \left[\sum_{i=1}^N a + \sum_{i=1}^N bx_i \right] = \frac{1}{N} \left[N \times a + b \times \sum_{i=1}^N x_i \right]$$

$$\frac{1}{N} \left[N \times a + b \times \sum_{i=1}^N x_i \right] = a + b \times \sum_{i=1}^N x_i$$

$$a + b \times \left(\frac{1}{N} \sum_{i=1}^N x_i \right) = a + b \times m(X)$$

$$\therefore m(a+bX) = a + b \times m(X)$$

2) Show that $\text{cov}(X, a+by) = b \times \text{cov}(X, y)$

$$\text{cov}(X, y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(y)), \text{ let } w = a + by$$

$$\text{cov}(X, w) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(w_i - m(w)) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(w_i - m(a + by))$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(w_i - m(a + by)) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a + by_i - (a + b \times m(y))) \text{ per Part 1}$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(b(y_i - m(y))) = b \times \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(y)) = b \times \text{cov}(X, y)$$

$$\therefore \text{cov}(X, a + by) = b \times \text{cov}(X, y)$$

3) Show that $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$, and in particular that $\text{cov}(X, X) = s^2$

part #1) $\text{cov}(X, y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(y)), \text{ let } w = a + bX$

$$\text{cov}(w, w) = \frac{1}{N} \sum_{i=1}^N (w_i - m(w))(w_i - m(w)) = \frac{1}{N} \sum_{i=1}^N (a + bx_i - m(a + bX))(a + bx_i - m(a + bX))$$

$$\frac{1}{N} \sum_{i=1}^N (a + bx_i - m(a + bX))(a + bx_i - m(a + bX)) = \frac{1}{N} \sum_{i=1}^N (a + bx_i - (a + b \times m(X)))(a + bx_i - (a + b \times m(X)))$$

$$\frac{1}{N} \sum_{i=1}^N (a + bx_i - (a + b \times m(X)))(a + bx_i - (a + b \times m(X))) = \frac{1}{N} \sum_{i=1}^N (bx_i - (b \times m(X)))(bx_i - (b \times m(X)))$$

$$\frac{1}{N} \sum_{i=1}^N (bx_i - (b \times m(X)))(bx_i - (b \times m(X))) = b \times b \times \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(x_i - m(X)) = b^2 \times \text{cov}(X, X)$$

$$\therefore \text{cov}(a + bX, a + bX) = b^2 \times \text{cov}(X, X)$$

Part #2) $\text{COV}(X, X) = S^2$, where $S(X) = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2}$

$$\text{COV}(X, X) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(X_i - m(X)) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2$$

$$S = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2}$$

$$S^2 = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2$$

$$\therefore \text{COV}(X, X) = S^2$$

4) Consider a non-decreasing transformation $g()$. Is it always true that $m(g(X)) = g(m(X))$?

Test #1: $g(X) = 2X$  $X = \{1, 2, 3\}$

$$m(g(X)) = \frac{1}{3} \sum_{i=1}^3 2X_i = \frac{1}{3}(2+4+6) = 4$$

$$g(m(X)) = \left(\frac{1}{3} \sum_{i=1}^3 X_i\right) \times 2 = \frac{1}{3}(1+2+3) \times 2 = 2+2 = 4$$

$$m(g(X)) = g(m(X))$$

Test #2: $g(X) = X^2$ when $X \geq 0$  $X = \{1, 2, 3\}$

$$m(g(X)) = \frac{1}{3} \sum_{i=1}^3 X_i^2 = \frac{1}{3}(1^2 + 2^2 + 3^2) = \frac{1}{3}(1+4+9) = \frac{14}{3}$$

$$g(m(X)) = g\left(\frac{1}{3} \sum_{i=1}^3 X_i\right) = \left[\frac{1}{3}(1+2+3)\right]^2 = \left[\frac{1}{3}(6)\right]^2 = 4$$

$$m(g(X)) \neq g(m(X))$$

I first chose a linear function to test, and $m(g(X)) = g(m(X))$ held up. Next, I just picked 3 X values on the function $g(X) = X^2$. $4 \neq \frac{14}{3}$, so the $m(g(X)) = g(m(X))$ isn't always true. Good thing I checked a non-linear transformation!

This makes sense logically because a non-linear, non-decreasing transformation such as $g(X) = X^2$ when $X \geq 0$ stretches the spacing between y values and the gaps don't stay the same distance apart. Since the mean is an average of values, it matters how far apart they are. A linear transformation could affect the size of these gaps, but in a consistent manner across all values in the function.